



LCD/Plasma TVs

Who needs to visit the theatres any more?

LCD or Plasma?

A difficult question: Plasma screens aren't generally smaller than 42 inches, so most people buying a TV smaller than that size needn't even consider Plasma as an option. A good Plasma panel will look better (sharper and crisper, with better contrast) than a good LCD panel. Plasma panels also have wider viewing angles. Conversely, even a mediocre LCD will look better than a *regular* Plasma. If you're going Plasma—make sure to stick with bigger brands—here's where you'll get all the visual quality—and the extra amounts you pay will be worth it considering the amounts bigger brands spend on research and development.

There's a debate about the life of Plasma panels. Suffice it to say it isn't a problem with most Plasma panels having a life of 40,000 hours, with the half-life of the phosphor gas in them being about 20,000 hours—that's nine years of happy viewing (assuming six hours a day)! An LCD will last you for 10,000 to 15,000 hours more. Plasma displays are still fairly new as a technology though, and improvements are being made even as you're reading this.



What You Should Be Looking At

Viewing Distance: When shopping for flat panels, remember to consider the viewing distance first. The bigger the panel, the bigger the dot pitch, and if you view a 50-inch screen from a distance of 12 feet, you'll wonder what all the hype is about—colours will appear to bleed, and the image will be aliased. Generally, for a 32-inch panel, we recommend a viewing distance of 15 feet—no less. For a 50-inch panel we'd recommend you sit at least 25 feet away. If your viewing distance is going to be less than 12 feet, look for a 26-inch.

Contrast and brightness: Look for a contrast ratio of at least 1500:1. The term *dynamic contrast ratio* (DCR) is

misleading—do not refer to this figure, and don't be fooled by manufacturers labelling 1000:1 static contrast ratio panel bearing TVs as having a DCR of 10000:1, or 6000:1. You want to see the static contrast ratio of a panel. Brightness is also important, and for movies, a brightness of 400 cd/m2 should suffice.

Resolution: A panel with a resolution of 1366 x 768 will also be stated as being HD ready. However, this is misleading, as this panel won't be able to display 1080p (1920 x 1080 pixels progressive scanned) content in all its glory, but will do 720p (1280 x 720)—which is why the manufacturer is *technically correct* when he claims support for "HD". Most panels in the market today sport a resolution of 1366 x 768, and although they can display higher resolutions the display isn't as good as a true 1080p panel. However, 1080p sources are rare, and such panels are astronomically priced. Most panels do however do 1080i (interlaced)—while this isn't as good as 1080p, 1080i will suffice in most cases.

Video Connectivity: Look for both D-Sub and DVI connectivity so your PC doesn't get left out. Component Video Inputs are *de facto*—this gives the best video quality amongst all the traditional inputs, and is far better than S-Video connectivity. Composite connectivity is another standard that is essential as an option, although mostly you'll get along with Component just fine. HDMI is a new standard and although newer panels are advertising HDMI support, please do look at the overall quality of the panel as the primary concern. HDMI is basically DVI + audio—nothing really great about it—although if your DVD player has an HDMI port then you'll get brilliant image + sound on a single connect. If however HDMI isn't mentioned (although that's rare nowadays) you'll manage with DVI—we reiterate, don't compromise on panel quality. ☐

If anyone told us three years ago that we'd have 55-plus-inch HDTVs in India, we'd have told them to take a hike. Three years later, we'd be swallowing our words! Large screen TVs have made it to the stage and we cannot wait for act two.

LCD / Plasma TVs are generally widescreen panels with support for at least 720p (1280 x 720 progressive scanned) content. Sizes range from 26 and 32 inches (LCDs) to larger panels up to 65 inches (both LCDs and Plasmas). However don't let the word "TV" fool you. Most of these panels can do much more than the mundane soap opera watching on cable TV channels. These TVs make great companions as screens for an immersive movie experience. For one they're much larger (as far as screen area goes) than a conventional CRT TV.

Most LCD TVs use PVA or S-PVA panels due to their higher contrast ratios—a boon for multimedia content. Although we have seen cheaper LCD TVs based on TN panels we suggest you give these a wide berth. For a PC monitor a TN panel may be acceptable, not when all you're doing is watching very contrast sensitive material like high definition movies! In Plasma screens, what we see are the "Multimedia Panels," capable of bright and vivid colours and contrasts.



Agent Tips

If you want 50 inches (or more), prefer Plasmas over LCDs. They're better technically and as far as viewing goes, totally superior to LCDs. Moreover, larger Plasmas are excellent value for money too.